

Vector Geometry, Calculus, Trigonometry

A set of questions curated by @icecreambobcat from the NSWMaths Discord

Question Credits:

- [NSW Mathematics](#) (Most solutions can be found here)
- @moksifu from the NSWMaths Discord
- @amiyuki7 from the NSWMaths Discord
- @magiicmushroom from the NSWMaths Discord
- @oneightseven from the NSWMaths Discord
- SGS Mathematics Extension 1 Internals 2025
- JRAHS Mathematics Extension 1 Trial 2022
- NSB Mathematics Extension 1 Trial 2023
- NSB Mathematics Extension 1 Trial 2024
- SGHS Mathematics Extension 1 Trial 2023

Question & Mark Distribution:

- Vectors – 7 questions; 24 marks
- Calculus – 14 questions; 65 marks
- Trigonometry – 5 questions; 21 marks

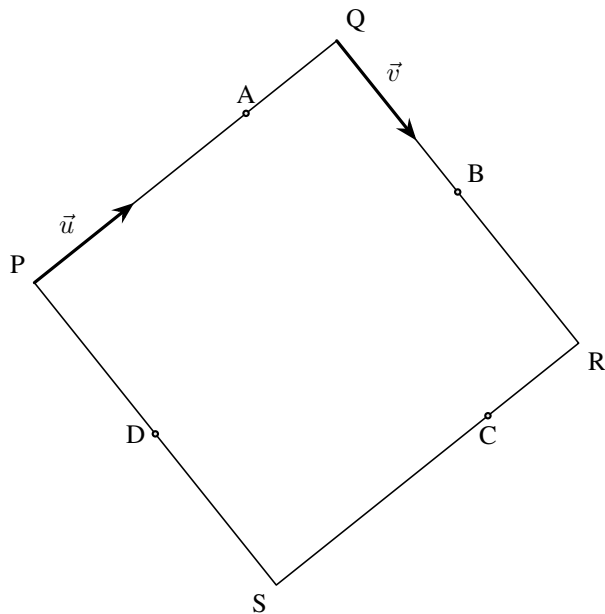
This is **NOT** intended to be completed under timed conditions; any timing guides are a reference only.

1 Vectors

Allow approximately 45 minutes for this section.

Question 1: Consider the points $A(-2, 12)$, $B(2, 8)$, $C(0, 1)$ and $D(-5, 0)$. There exists a unique square $PQRS$ such that each of the four points A , B , C and D lie on a different side of $PQRS$.

The **unit** vector in the direction of $\overrightarrow{PQ}$ is $\underline{u} = \begin{bmatrix} p \\ q \end{bmatrix}$, where q and p are positive, and the **unit** vector in the direction of $\overrightarrow{QR}$ is $\underline{v}$, as shown in the diagram below.



(i) Write down an expression for $\underline{v}$ in terms of p and q .

1

(ii) Hence, calculate the exact area of $PQRS$.

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Question 2: Define a sequence of vectors given by $T_0 = \mathbf{v}$ and T_n is the vector from projecting T_{n-1} onto $\mathbf{u}$. Prove by mathematical induction that for $n \geq 1$,

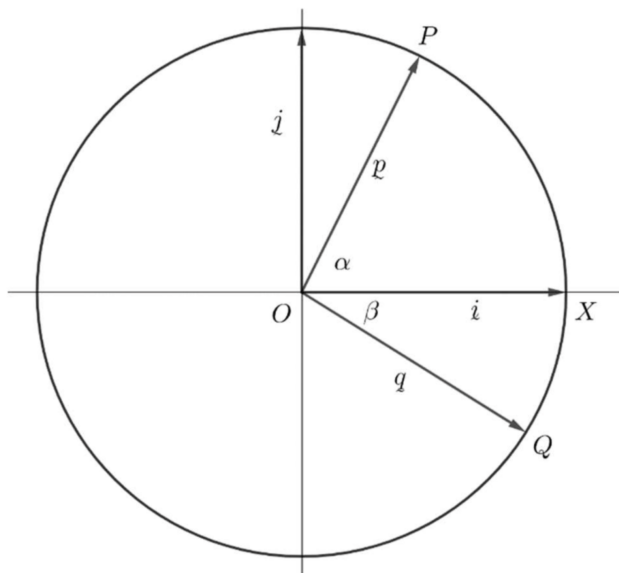
$$T_n = \text{proj}_{\mathbf{u}} \mathbf{v}.$$

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Question 3: In triangle OAB , M is the midpoint of AB and P internally divides OB in the ratio $3 : 1$. Q is the point on AP such that O , Q and M are collinear. Find the ratio $OQ : QM$.

2

Question 4: Consider the unit circle around the origin O in the diagram below, with $\underline{i}$ and $\underline{j}$ being the standard basis unit vectors in the horizontal and vertical directions respectively. The points P and Q lie on the unit circle such that P is in the first quadrant and Q is in the fourth quadrant. The angles POX and QOX have measures α and β respectively, where X is the point $(1, 0)$. Let $\overrightarrow{OP} = \underline{p}$, $\overrightarrow{OQ} = \underline{q}$.



(i) Find $\underline{p}$ in terms of α , and $\underline{q}$ in terms of β .

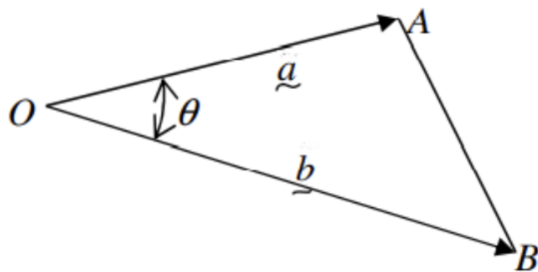
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(ii) Hence, by considering $\underline{p} \cdot \underline{q}$, show that

$$\cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta)$$

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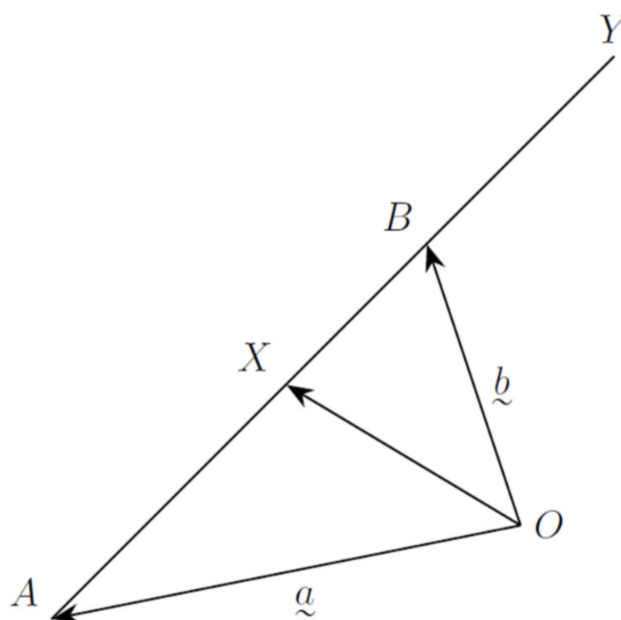
Question 5: Triangle OAB is defined by two non-parallel and nonzero vectors $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$. The angle between the vectors is θ .



Express the area of triangle OAB in terms of dot products of $\underline{a}$ and $\underline{b}$.

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Question 6: The point X is chosen within the interval AB such that $AX : XB = m : n$, where m and n are positive and $m \neq n$. Let $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$, as shown.



(i) Show that $\overrightarrow{OX} = \frac{n\underline{a} + m\underline{b}}{m + n}$.

2

(ii) The point Y is on the line AB such that

$$\overrightarrow{OY} = \frac{n\underline{a} - m\underline{b}}{m - n}.$$

(Do **not** prove this)

If OX is perpendicular to OY , show that $n|\underline{a}| = m|\underline{b}|$.

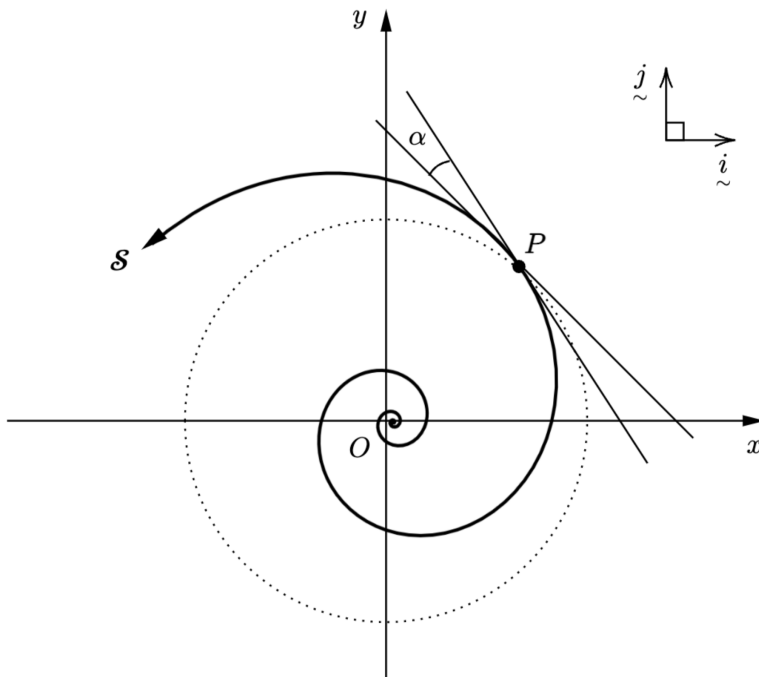
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Question 7: The spiral $\mathcal{S}$ is defined by the parametric vector

$$\underline{r}(\theta) = \begin{pmatrix} ae^{k\theta} \cos \theta \\ ae^{k\theta} \sin \theta \end{pmatrix}$$

for $a, k \in \mathbb{R}^+$.

The *pitch angle*, α , at any point P on $\mathcal{S}$ is the angle between the tangent to the circle centered at O with radius $|\overrightarrow{OP}|$, and the tangent to $\mathcal{S}$ at P .



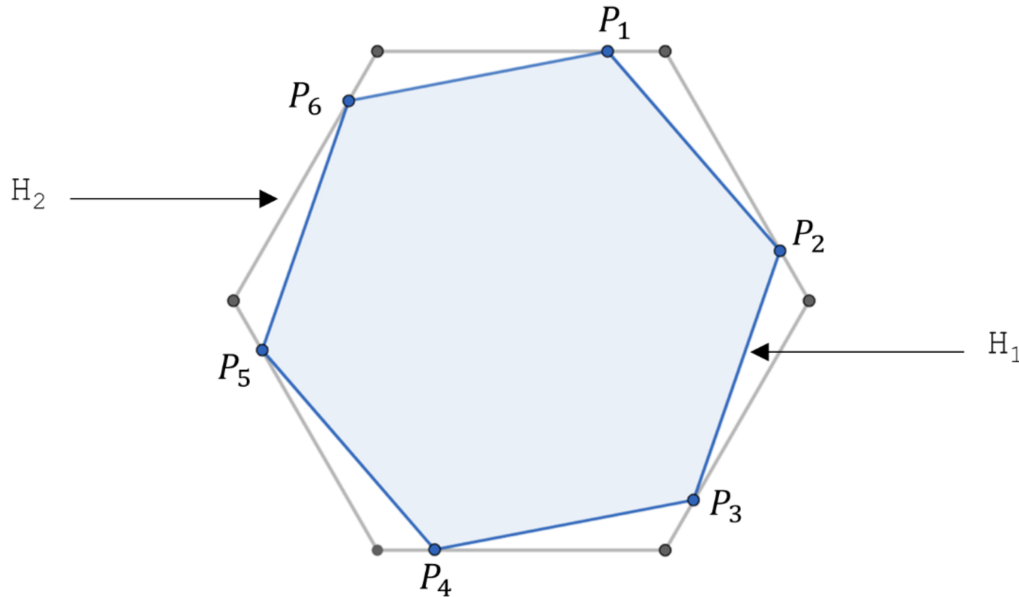
Prove $\tan \alpha = k$.

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2 Calculus

Allow approximately 120 minutes for this section.

Question 1: Six points P_1, P_2, P_3, P_4, P_5 and P_6 form the vertices of a regular hexagon H_1 . An animated program is rotating these six points at a constant speed of 3 cm/s around the perimeter of another regular hexagon, H_2 , which has a side length of 20cm.



Initially, the six vertices of H_1 lie on top of the six vertices of H_2 . Determine the rate of change of the area H_1 ten seconds after the animation begins. 4

Question 2: Consider the quadratic function $f(x) = px^2 + qx + r$.

The points $A(a, f(a))$ and $B(b, f(b))$ lie on the graph $y = f(x)$. The secant line ℓ_S passes through A and B . It is known that there exists a unique line ℓ_T that is parallel to ℓ_S , and tangent to the graph $y = f(x)$ at the point $C(c, f(c))$. (Do **not** prove the existence of ℓ_T).

(i) Prove that $c = \frac{a+b}{2}$. 1

(ii) The lines $\ell_S, \ell_T, x = a$, and $x = b$ enclose a parallelogram region. Prove that the area of this region is

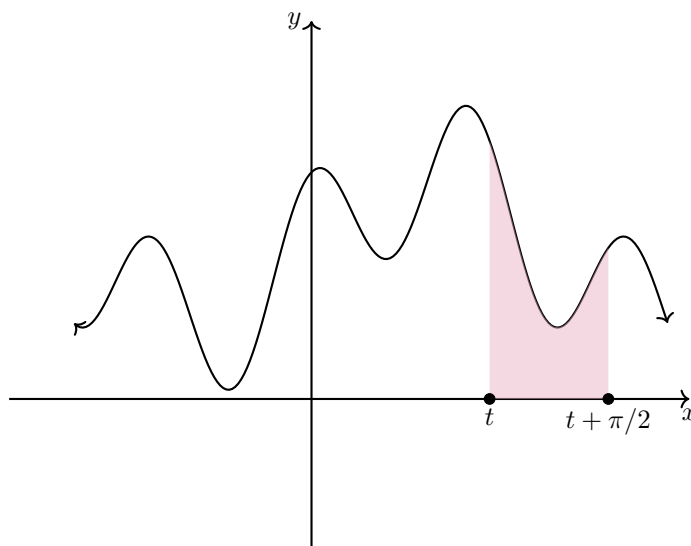
$$\left| \frac{p(a-b)^3}{4} \right|.$$

3

(iii) Hence, or otherwise, prove that the area of $\triangle ABC$ is $\frac{3}{4}$ of the area enclosed by ℓ_S and the graph $y = f(x)$. 4

Question 3: The graph of $f(x) = 2 + \sin x + \cos 3x$ is shown below.

Let $g(t)$ be the area under the graph $y = f(x)$ on the interval $\left[t, t + \frac{\pi}{2}\right]$.



(i) Show that

$$g(t) = \pi + \sqrt{2} \left(\sin \left(t + \frac{\pi}{4} \right) - \frac{1}{3} \sin \left(3t + \frac{\pi}{4} \right) \right).$$

3

(ii) Hence, find the exact maximum area of a $\frac{\pi}{2}$ -wide slice.

3

Question 4: Show, using calculus, that for $x > 0$,

$$\ln(1+x) - x + \frac{x^2}{2} < \frac{x^3}{3}.$$

3

Question 5: Let $p(x)$, $q(x)$ and $r(x)$ be polynomials. Let

$$F(x) = \left(\int_1^x p(t) dt \right) \left(\int_1^x q(t) dt \right) \left(\int_1^x r(t) dt \right).$$

Show that if the degree of $F(x)$ is at least 3, then $(x-1)^3$ is a factor of $F(x)$.

4

Question 6: Suppose m and n are non-negative integers such that $m < n$. Using the substitution $u = \tan x$, show that

$$\int_0^{\frac{\pi}{4}} \frac{\sin^{2m} x}{\cos^{2n} x} dx = \sum_{k=0}^{n-m-1} \binom{n-m-1}{k} \frac{1}{2k+2m+1}.$$

4

Question 7: A differentiable function f satisfies the equation

$$f(x) = x \int_1^x f(t) dt.$$

(i) Explain why $f(1) = 0$.

1

(ii) Find all f .

2

Question 8: Given

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}},$$

(i) Show that $f(x)$ has no stationary points.

2

(ii) Given that $y = \pm 1$ are horizontal asymptotes, sketch the curve.

2

(iii) For $k > 0$, consider the area enclosed by the curve, the lines $y = 1$, $x = 0$, and $x = k$. Show that this area can be expressed in the form

$$\ln \left(\frac{2e^k}{e^k + e^{-k}} \right).$$

2

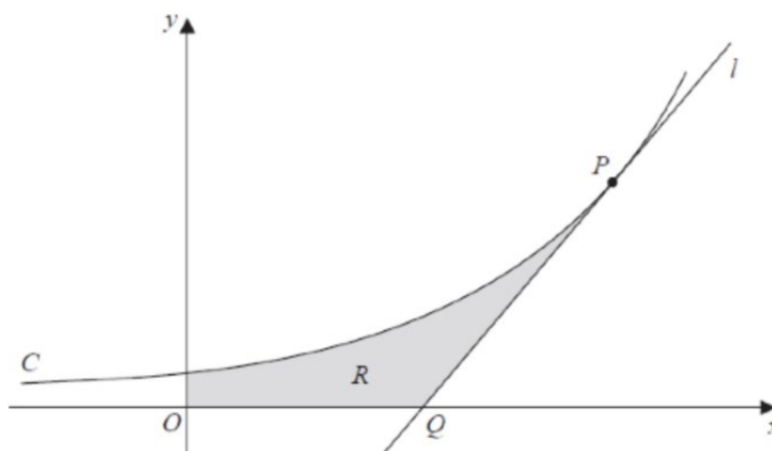
(iv) Hence, justify why for all values of k , the area found in part (iii) is always less than 2.

1

Question 9: By using the substitution $u = \tan^{-1} x$, evaluate $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$.

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Question 10: The diagram shows a region bound by the curve C with the equation $y = 3^x$, the line l , and the x -axis. The x coordinates of points P and Q are 2 and $\left(2 - \frac{1}{\ln 3}\right)$ respectively.



A solid is created when the region R is revolved around the x axis. Find the volume of the solid formed.

4

Question 11: A hemi-spherical bowl has a radius of 3m. Oil is poured in at a constant rate of $\frac{\pi}{3}\text{m}^3/\text{min}$.

- (i) Show that when the depth of the oil is h metres, the volume of oil is given by

$$V = \frac{\pi}{3}(9h^2 - h^3).$$

2

- (ii) Show that the depth of the oil in the bowl after 8 minutes is 1 meter.

3

- (iii) At what rate is h increasing at this time?

2

Question 12:

- (i) By equating coefficients, find the values of P and Q such that

$$P(2 \sin x + \cos x) + Q(2 \cos x - \sin x) \equiv 7 \sin x + 11 \cos x.$$

2

- (ii) Hence, or otherwise, evaluate

$$\int_0^{\frac{\pi}{2}} \frac{7 \sin x + 11 \cos x}{2 \sin x + \cos x} dx.$$

2

Question 13: Evaluate:

- (i)

$$\int \frac{\sin x}{\sin x + \sec x} dx.$$

3

- (ii)

$$\int \frac{dx}{2 + \tan^2 x}.$$

2

Question 14: Show that

$$\int_{-\frac{6}{5}}^{-\frac{2}{5}} \frac{\sqrt{25x^2 - 4}}{x} dx = \pi - 4\sqrt{2} - 2 \sin^{-1} \left(\frac{1}{3} \right).$$

3

3 Trigonometry

Allow approximately 45 minutes for this section.

Question 1: Consider the equation

$$\frac{\sqrt{3}-1}{\sin x} + \frac{\sqrt{3}+1}{\cos x} = 4\sqrt{2}$$

for $0 < x < \frac{\pi}{2}$. It is given that $\sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}-\sqrt{2}}{4}$ and $\cos\left(\frac{\pi}{12}\right) = \frac{\sqrt{6}+\sqrt{2}}{4}$.

(i) Verify that $x = \frac{\pi}{12}$ is a solution to the equation.

1

(ii) Hence, find the other solution to the equation for $0 < x < \frac{\pi}{2}$.

3

Question 2: Consider the polynomial equation $x^3 - 5x^2 - 12x + 8 = 0$.

(i) Show that the equation has exactly one negative root, and two positive roots.

2

(ii) Let the roots of the equation be $\tan \alpha$, $\tan \beta$ and $\tan \gamma$, where $\alpha, \beta, \gamma \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Justify, with appropriate calculations and reasoning, why

$$\alpha + \beta + \gamma = \frac{\pi}{4}.$$

3

Question 3: Solve the following equation for x , giving your answer as exact values:

$$\sin\left(2\cos^{-1}\left(\cot\left(2\tan^{-1}(x)\right)\right)\right) = 0.$$

2

Question 4: Let θ be the measure of an acute angle.

(i) Using a suitable expansion of $\sin 6\theta$, show that

$$(\sin 2\theta)^3 - \frac{3}{4}\sin 2\theta + \frac{1}{4}\sin 6\theta = 0.$$

2

(ii) If $x = 4\sin 2\theta$ and $x^3 - 12x + 8 = 0$, show that $\sin 6\theta = \frac{1}{2}$.

1

(iii) Using the result in (ii), evaluate

$$\left(\sin \frac{\pi}{18}\right)^2 + \left(\sin \frac{13\pi}{18}\right)^2 + \left(\sin \frac{25\pi}{18}\right)^2$$

3

Question 5: By considering an appropriate t-formula substitution, solve

$$3\cos\left(\frac{\theta}{2}\right) - \sin\left(\frac{\theta}{2}\right) = -3$$

for $\theta \in [0, 2\pi]$.

4